Understanding Student Engagement with Live Music: A Data-Driven Analysis of Concert Attendance at Pomona College's Music Department

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Introduction

For the duration of my Capstone project, I am working with Professor Joti Rockwell and Concert Production Manager Audrey Dunne to explore the multifaceted factors that influence student attendance at concerts hosted by the Pomona College Music Department. This investigation combines rigorous historical data analysis spanning 15 years with active, hands-on collection of contemporary engagement metrics to develop a comprehensive understanding of attendance patterns, their underlying causes, and their implications for strategic programming decisions. The project addresses a central challenge facing arts organizations across higher education: how to cultivate sustained student engagement with live performance amid competing demands, digital entertainment alternatives, and evolving campus cultures.

The music department's concert series represents a significant institutional investment in cultural programming, offering students regular opportunities to experience live music across diverse genres and performance contexts. Yet attendance patterns remain inconsistent and poorly understood, making it difficult for the department to optimize scheduling, allocate promotional resources effectively, or design programming that resonates with student interests and availability. By applying systematic data analysis to this question, this project aims to transform attendance from an unpredictable variable into a measurable, interpretable phenomenon that can inform evidence-based decision-making.

I have structured this investigation into three interconnected temporal phases—Past, Present, and Future—each designed to address distinct research questions while contributing to an integrated understanding of concert attendance dynamics. This tripartite framework allows me to trace how historical events and long-term trends have shaped current patterns, document real-time engagement behaviors and motivations, and project forward to develop sustainable strategies for enhancing participation. The three sections are not merely chronological divisions but represent different methodological approaches and analytical lenses, each offering unique insights that collectively paint a complete picture of the factors influencing student attendance decisions.

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Fall 2025 · Fall 2026

Past

Analysis of data (attendance logs for concerts) collected over the past 15 years.

Research Questions

- **How did COVID-19 impact concert attendance?**
- Do different concert types (classical, contemporary, student vs. faculty performances) show distinct attendance patterns?
    - How should I categorize the concerts?
- Facult/guest concerts / composers / genres / instruments/ weekday
- How did changes in student body composition (size, demographics, academic programs) relate to attendance patterns?
    - Meeting with Dean Arwari
- How do other campus activities (including academic) impact attendance?
    - Scrape engage?

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Present

Collecting new data via social media and surveys.
Instagram | Bitly | Google dashboard | Facebook (idk if I got invite link)

Research Questions

Data Collection

- What entices students, specifically, to attend a concert?
    - Known factor: their friend is playing in the concert! If that's not relevant, e.g. faculty or guest performances, what other factors contribute to their interest?
- What are the primary barriers preventing interested students from attending? (time, location, awareness, perceived relevance)
    - All events in the past on Engage/Pomona calendar
- **How students perceive concert experience as part of their curriculm?**
    - Post vs pre concert survey
- What did you learn from/like about the concert?
- Experience on a scale from 1-10?
- **Gauging enthusiasm?**

Concert Attendee Survey Link
- Pre concert vs post concert

General Feedback Survey Link
- Chirps + feedbacker?

Social Media Analytics

- What posts generate the most engagement, and how does this translate to attendance? Why?
- What is the optimal timing and frequency for concert promotion posts? Does this vary per platform?
- How does the visual content type (flyers, stills, behind-the-scenes content) affect engagement rates?
- What role do student vs. official account endorsements play in driving attendance?

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Fall 2025 · Fall 2026

Present

While historical data illuminates long-term trends and structural factors, it cannot reveal the immediate motivations, barriers, and decision-making processes that shape individual students' choices about concert attendance. The second phase of my project addresses this limitation by actively collecting contemporary data through student surveys and social media analytics conducted during Fall 2025. This present-focused investigation captures the current landscape of student engagement, documenting both quantitative metrics of digital and in-person participation and qualitative insights into students' subjective experiences and perspectives.

Survey methodology forms a cornerstone of this phase. In collaboration with Professor Rockwell and Audrey Dunne, I developed a post-concert feedback survey in Qualtrics, distributed via QR codes displayed on concert screens and printed in program materials. The survey includes both quantitative rating scales and multiple-choice questions designed to capture diverse dimensions of the concert experience: whether respondents are first-time or returning attendees; what factors motivated their decision to attend; how they learned about the concert; their institutional affiliation (Pomona student, other 5C student, faculty/staff, community member); their satisfaction level on a 0-100 scale; whether they attended alone or with others; and their likelihood of attending future concerts. Through early data collection spanning four concerts between September and October 2025, I have already gathered 65 responses that reveal surprising patterns—most notably, that the majority of concert attendees (55%) are non-students from the broader Claremont community, while Pomona students represent only 22% of the audience. This finding has profound implications for understanding the "student attendance problem" and designing appropriate interventions.

The survey data reveals additional insights that challenge assumptions and highlight opportunities. An overwhelming 93% of respondents reported attending previous Music Department concerts, indicating strong retention among the existing audience but raising questions about the department's ability to attract newcomers. Satisfaction levels are exceptionally high (mean score of 93.69 out of 100), suggesting that quality and programming are not the primary obstacles to student attendance. When asked what factors influenced their decision to attend, respondents most frequently cited enjoyment of concerts generally (63%), interest in the type of music or performance (43%), and knowing a performer (39%). Importance ratings for various decision factors revealed that genre/type of performance (mean: 71.69),

location (mean: 67.40), and timing (mean: 65.42) ranked as most influential, underscoring the significance of programming choices and logistical considerations. Interestingly, traditional printed materials (flyers, posters, programs) proved most effective at reaching this audience (60%), followed by word of mouth (37%) and social media (34%), likely reflecting the predominance of community members who may be less engaged with digital platforms than student audiences.

Complementing these survey findings, I am conducting a systematic analysis of social media analytics to understand how digital engagement relates to physical attendance. By downloading .csv files from Facebook, Instagram, Google Dashboard, and Bitly—platforms the department uses to promote concerts—I am tracking metrics such as post reach, impressions, likes, comments, shares, story views, link clicks, and profile visits. Visualizing these data streams alongside attendance figures allows me to assess which content types generate the most meaningful engagement, whether increased digital visibility translates into higher turnout, and how different platforms vary in their effectiveness. For example, does behind-the-scenes rehearsal footage generate more interest than formal promotional posters? Do posts featuring student performers receive higher engagement from the student demographic? Is there an optimal timing and frequency for concert promotion posts, or does excessive posting lead to diminishing returns? These questions will guide the department's social media strategy and help allocate limited promotional resources more effectively.

The present phase also involves ongoing refinement of data collection instruments based on early feedback. As I continue surveying audiences at subsequent concerts, I am incorporating suggestions from respondents and stakeholders to improve question clarity, expand demographic categories, and better capture the student experience specifically. This iterative approach ensures that the data I collect remains relevant, actionable, and responsive to the project's evolving understanding of attendance dynamics.

Breakdown of Survey Responses

First-Time vs. Returning Attendees

The overwhelming majority of respondents were not first-time attendees. This distribution indicates that the Music Department successfully retains an engaged core audience who attend concerts repeatedly. 93 percent of attendees had attended a Music Department concert before. However, the very low proportion of first-timers (only 7%) raises questions about whether the department is effectively reaching new audiences or whether concerts primarily serve an established, loyal base.

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Fall 2025 · Fall 2026

Institutional Affiliation

- Non-5C community members (parents, community members, employees): 55%
- Pomona College students: 22%
- Other 5C students (parents, employees, CMC, Scripps, Harvey Mudd, Pitzer): Combined ~23%

Rather than being primarily a student-oriented activity, Music Department concerts draw a majority of their audience from the broader Claremont community. Parents, local residents, and college employees make up over half the audience, with Pomona students representing less than a quarter of attendees. This has significant implications for promotional strategy, programming decisions, and assessment of "student attendance" as a metric.

Music Major Status

- Music majors: Approximately 4%
- Non-music majors: Approximately 30%
- Undecided majors: Approximately 1%
- Non-students: Approximately 64%

Even among student respondents, the overwhelming majority are not music majors, indicating that concerts successfully attract beyond the department for those students who do attend.

How Attendees Heard About Concerts

- Flyers/Printed Materials: 60%
- Word-of-Mouth: 37%
- Social Media: 34%

Traditional printed materials prove most effective for reaching this audience, which makes sense given the 55% community (non-student) demographic. These community members may be less engaged with social media platforms and more responsive to physical signage and printed programs.

However, this also reveals a potential disconnect: if outreach efforts focus heavily on digital platforms popular with students (Instagram, Facebook), they may be optimized for the wrong primary audience.

What Brought Returning Attendees Back

- "I just enjoy concerts!": 63%
- Interest in type of music/performance: 43%
- Knowing a performer: 39%

This suggests that the established audience attends consistently but still responds to programming choices. For this loyal base, variety and interesting repertoire selections help maintain engagement even when they don't know performers personally.

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Importance of Decision Factors

Respondents rated various factors on a 0-100 scale for how important each was in their decision to attend. The top factors were:

1. Genre of Music/Type of Performance: Mean = 71.69
2. Location: Mean = 67.40
3. Timing: Mean = 65.42
4. General Curiosity: Mean = 52.52

Programming decisions emerge as the most consistently important factor. Combined with Q2 data showing 43% motivated by "interest in type of music/performance," this suggests repertoire selection is a powerful lever for both attracting and retaining audiences.

Attending Alone vs. With Others

- Attending with others: 66%
- Attending alone: 34%

Here, we see the role that relationships could play in concert attendance. This distribution likely reflects the mixed audience: students may be more likely to attend in groups, while community members might attend independently or with one other person, or vice versa.

Overall Satisfaction Level

Respondents rated their concert experience on a 0-100 slider scale:

- Mean: 93.69
- Standard Deviation: 7.01
- Range: 73-100
- Total Responses: 65

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Fall 2025 · Fall 2026

Future

Brainstorming ways to drive engagement based on data.

Research Questions

- How can we get newcomers to concerts?
- How can we get first-time attendees to return to another concert?
- What columns can we add to the dataset to acquire more information in the future?
- What post-concert feedback mechanisms would provide actionable insights?
- **Can we develop models to predict attendance based on historical patterns, social media engagement, and campus factors?**

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Capstone in Data Science
Fall 2025 · Fall 2026

Bibliography

This article examines the concert attendance behaviors of undergraduate non-music majors, focusing on how preferences, convenience, and course requirements influence participation. Through three studies, Flowers analyzes correlations between students' stated musical preferences and actual attendance, the timing of attendance in relation to assignment deadlines, and the vocabulary students use in describing concerts. The findings highlight discrepancies between intention and behavior, and show that convenience often outweighs preference in determining attendance choices. This source will provide valuable context for understanding how student motivations and institutional factors shape attendance patterns. It will help me interpret Pomona's historical attendance logs and survey results, and extrapolate deliverable suggestions from my analysis.

Flowers, Patricia J. "ASPECTS OF CONCERT ATTENDANCE BEHAVIOR OF UNDERGRADUATE NONMUSIC MAJORS." *Contributions to Music Education*, no. 10, 1983, pp. 19–26. *JSTOR*, http://www.jstor.org/stable/24127374. Accessed 10 Sept. 2025.

This case study examines how the Emily Carr University Library utilized Facebook, Instagram, and Twitter to engage students and community members, assessing effectiveness through analytics and surveys. The authors discuss strategies for increasing visibility, the importance of goal setting, and the challenges of measuring success beyond simple metrics, such as follower counts. A key finding is that Instagram provided the highest engagement due to its visual nature, with social media activity influencing not only awareness but also physical attendance at library events. This article will support the "Present" phase of my capstone project by offering concrete examples of how social media platforms can drive participation in arts-related events. It will also

help guide my analysis of Instagram and Bitly data, particularly in connecting online engagement with concert attendance outcomes.

Kulczynski, Alicia, et al. “Measuring Motivations for Popular Music Concert Attendance.” *Event Management*, vol. 20, no. 2, 2016, pp. 239–54, https://doi.org/10.3727/152599516X14643674421816.

This dataset documents attendance at music department concerts over a 15-year period, including student, faculty, and guest performances. It will serve as the foundation for the “Past” phase of my capstone project, allowing me to identify long-term patterns and the effects of major events such as COVID-19 on participation. By combining these logs with demographic and campus activity data, I will analyze how concert type, timing, and student body changes influence attendance trends.

Pomona College Music Department. (2008–2023). Concert attendance logs [Unpublished internal dataset]. Pomona College.

This case study evaluates how the Emily Carr University Library used Facebook, Instagram, and Twitter to engage students and community members, measuring effectiveness through analytics and surveys. The authors discuss strategies for increasing visibility, the importance of goal-setting, and the challenges of measuring success beyond simple metrics like follower counts. A key finding is that Instagram provided the highest engagement due to its visual nature, with social media activity influencing not only awareness but also physical attendance at library events. This article will support the “Present” phase of my capstone project by offering concrete examples of how social media platforms can drive participation in arts-related events. It will also help guide my analysis of Instagram and Bitly data, particularly in connecting online engagement with concert attendance outcomes.

Solomon, Mary, "Market Research on Student Concert Attendance at BGSU's College of Musical Arts" (2019). Honors Projects. 440. https://scholarworks.bgsu.edu/honorsprojects/440

Jazzmin Duncan
Capstone in Data Science
Fall 2025 · Fall 2026

This study develops and validates the Concert Attendance Motivation Scale (CAMS), designed to measure the diverse motivations that influence popular music concert attendance. Using focus groups, exploratory factor analysis, and survey data from over 500 participants, the authors identify ten key motivational dimensions, including nostalgia, aesthetics, escape, social interaction, hero worship, and uninhibited behavior. Three motivations—nostalgia, status enhancement, and the opportunity to experience new or concert-specific music—are highlighted as unique to popular music contexts. The paper demonstrates how these motivational categories can inform marketing and event management strategies. For my capstone project, this source provides a comprehensive framework for understanding the psychological and social drivers of concert attendance, which I can apply when analyzing survey results and considering long-term strategies to encourage student participation in Pomona's music events.

Webb, Hillary, and Ken Laing. "Engaging with Social Media: The Emily Carr University of Art and Design Library Experience." *Art Documentation: Journal of the Art Libraries Society of North America*, vol. 34, no. 1, 2015, pp. 137–51. *JSTOR*, https://doi.org/10.1086/680570. Accessed 10 Sept. 2025.

Jazzmin Duncan
Capstone in Data Science
Fall 2025 · Fall 2026

Outline #1

I. Introduction

This section establishes the context and scope of the project, introducing the partnership with the Music Department, specifically with Professor Joti Rockwell and Concert Production Manager Audrey Dunne. I'll explain the structure (Past, Present, Future) and articulate the overarching goal: to understand what drives student attendance at music department concerts and develop data-informed strategies for sustained engagement. I'll also preview the key datasets being used and briefly outline how these sources will inform actionable recommendations for the music department.

II. Literature Review

This section will synthesize existing research on concert attendance behaviors and engagement strategies based on the sources I've collected and will collect. This review will establish theoretical grounding for my methodology and analysis.

III. Past: Historical Data Analysis (2008–2023)

This section presents findings from 15 years of concert attendance logs. I'll begin with descriptive statistics and trend visualizations showing attendance patterns over time, considering changes in concert formats (virtual vs. in-person), concert types (classical, contemporary, student performances, faculty recitals, guest artists), and student body demographic data from ASPC (assuming I'm given access).

Jazzmin Duncan
Capstone in Data Science
Fall 2025 · Fall 2026

IV. Present: Contemporary Data Collection and Analysis

This section documents current data collection efforts and preliminary findings. I'll describe the survey methodology, including pre-concert and post-concert questionnaires designed to capture student motivations, barriers to attendance (time, awareness, location, perceived relevance), and concert experience ratings. Social media analytics from Instagram, Facebook, Google Dashboard, and Bitly will be visualized to reveal which content types (flyers, behind-the-scenes footage, student testimonials) generate the most engagement and how digital interactions correlate with actual attendance.

V. Future: Predictive Modeling and Strategic Recommendations

Building on insights from historical and contemporary data, this section proposes reproducible strategies for enhancing concert attendance. I'll recommend new variables to track longitudinally (music major status, general campus event participation rates, extracurricular music engagement) and suggest post-concert feedback mechanisms (brief rating scales, qualitative response prompts) that balance data richness with student burden. The centerpiece will be a predictive model that estimates attendance based on concert type, historical trends, social media engagement metrics, academic calendar positioning, and competing campus activities.

VI. Conclusion

This final section will synthesize key findings across all three temporal phases, reflecting on what the data reveals about the complex factors shaping student concert attendance. I'll discuss limitations of the study, including potential gaps in historical data and the challenges of isolating causal factors in a dynamic campus environment. I'll emphasize the project's practical contributions to the music department's strategic planning and suggest directions for future research.

VII. Appendices

- Survey instruments (pre-concert and post-concert questionnaires)
- Social media analytics dashboard samples
- Data visualization examples
- Statistical model specifications and validation metrics

Jazzmin Duncan
Capstone in Data Science
Fall 2025 · Fall 2026

I. Introduction

Live music occupies an important place in the cultural and educational landscape of higher education. However, maintaining an engaged audience presents a persistent challenge for institutions investing in live performance programming. Students must navigate competing academic responsibilities, extracurricular commitments, social activities, and increasingly accessible forms of digital entertainment. Consequently, even when institutions offer regular opportunities to experience live music, student attendance is not necessarily consistent or predictable.

This capstone project investigates the multifaceted factors influencing student attendance at concerts hosted by Pomona College's Music Department. Working in collaboration with Professor Joti Rockwell and Concert Production Manager Audrey Dunne, I examine historical attendance records, contemporary audience engagement, and potential strategies for improving participation. By combining 16 years of attendance data with surveys and digital engagement metrics, this project seeks to develop a more comprehensive understanding of how students discover, evaluate, and ultimately decide whether to attend live musical performances.

Although an individual's interest in a particular genre or performer may influence their willingness to attend, other considerations—including scheduling, location, social relationships, awareness of upcoming events, and competing campus activities—may be equally consequential. A student might express enthusiasm for a performance but ultimately decide not to attend because of an inconvenient schedule or a lack of awareness about the event. Conversely, students who are unfamiliar with a particular genre might attend because a friend is performing or because they encounter an appealing promotional campaign. Understanding these distinctions requires examining both the motivations that encourage attendance and the structural or situational barriers that discourage it.

To investigate these questions, I have organized the project into three interconnected phases: Past, Present, and Future. These phases provide a chronological framework while also representing distinct methodological approaches to understanding audience engagement.

The first phase, *Past: Historical Data Analysis*, examines approximately 16 academic years of concert attendance records, spanning 2008–2026. By analyzing historical attendance patterns, I aim to identify changes in participation over time and investigate how institutional and external circumstances may have influenced concert attendance. Particular attention will be given to the effects of the COVID-19 pandemic, differences between performance types, and potential relationships between attendance and changes in the student population. This historical

perspective establishes a baseline for understanding the department's audience and provides context for interpreting more recent engagement patterns.

The second phase, *Present: Contemporary Data Collection and Analysis*, focuses on newer information gathered about audience motivations and promotional effectiveness. This component combines audience surveys with social media and digital engagement analytics. Surveys distributed through Qualtrics provide insight into attendees' institutional affiliations, reasons for attending, sources of event awareness, satisfaction levels, and likelihood of returning to future performances. Meanwhile, analytics from platforms including Instagram, Facebook, Google Dashboard, and Bitly allow me to examine how audiences interact with promotional materials before concerts take place. Together, these datasets help illuminate the relationship between digital visibility, audience interest, and in-person participation.

Future: Predictive Modeling and Strategic Recommendations builds on the historical and contemporary findings to identify opportunities for improving student participation. By integrating attendance records, survey responses, digital engagement metrics, and information about the academic calendar and competing campus activities, I hope to create a Shiny application for the Department to use moving forward.

Ultimately, this project seeks to bridge the gap between data collection and practical application, and to provide a reproducible process of understanding audiences moving forward. By examining the relationship between programming, accessibility, promotion, and student engagement, this investigation aims to provide the Pomona College Music Department with an evidence-based, interactive framework for understanding its audiences and strengthening its connection with the campus community.

Jazzmin Duncan
Capstone in Data Science
Fall 2025 · Fall 2026

Ethical Considerations

Because this project involves collecting information from concert attendees, examining audience engagement, and analyzing institutional records, several ethical considerations have informed my research process. These include considering respondent privacy, weighing bias in creating data categories and survey responses, maintaining the confidentiality of institutional data, interpreting findings responsibly, and recognizing the limitations of using digital engagement metrics to conclude individual behavior.

The primary ethical concerns rise throughout the survey process. Although the surveys are designed to understand audience motivations, experiences, and satisfaction, respondents should not feel pressured to participate or disclose personal information. The survey questions therefore focus primarily on their concert experiences, institutional affiliations, sources of event awareness, and likelihood of attending future performances rather than collecting unnecessary identifying information. Additionally, survey findings are presented in aggregate form, allowing broader audience trends to be examined without unnecessarily exposing individual respondents' answers.

Another consideration involves the handling of historical attendance records and institutional data. Since these records document participation in college-sponsored events, they contain some information that directly identifies individual students or attendees. Such information must be treated responsibly, particularly when combining attendance records with other datasets. Any future integration of demographic or institutional datasets must also account for privacy considerations and appropriate access permissions.

The use of social media analytics presents another ethical challenge. Digital engagement metrics, including impressions, likes, shares, profile visits, and link clicks, provide insight into how audiences interact with promotional content, but they do not necessarily establish an individual's interest in or intention to attend a concert. Similarly, publicly visible engagement should not be treated as a direct representation of student opinion or overall audience sentiment. To address this limitation, I interpret digital engagement alongside survey responses and actual attendance figures rather than treating any single metric as a definitive measure of audience interest.

Finally, this project recognizes the ethical implications of translating quantitative findings into institutional recommendations. Attendance patterns may reflect circumstances beyond students' control, including academic workloads, scheduling conflicts, accessibility, and competing responsibilities. Consequently, low attendance should not automatically be interpreted as a lack of interest in the Music Department's programming. Instead, the project considers how institutional practices, promotional strategies, and logistical barriers may influence participation. The goal is not to assign responsibility to students for low attendance, but to identify

opportunities for the department to make its programming more accessible, visible, and engaging.

Ultimately, these considerations reinforce the importance of approaching audience research with transparency, confidentiality, and care. By prioritizing aggregate analysis, limiting unnecessary personal information, acknowledging methodological limitations, and contextualizing findings within the broader student experience, this project aims to produce recommendations that are both analytically meaningful and ethically responsible.

Jazzmin Duncan
Capstone in Data Science
Fall 2025 · Fall 2026

II. Literature Review

This section synthesizes existing scholarship on concert attendance behavior, audience motivations, and digital engagement strategies. It establishes the theoretical and methodological foundation for interpreting the project's historical attendance records, survey findings, and social media analytics.

III. Past: Historical Data Analysis (2008–2023)

This section examines approximately fifteen years of Pomona College Music Department concert attendance records. It investigates long-term attendance patterns, the effects of COVID-19, differences across concert types, and potential relationships between student population changes, academic scheduling, and audience participation.

IV. Present: Contemporary Data Collection and Analysis

This section documents the collection and analysis of contemporary audience survey responses and digital engagement metrics. It examines attendee demographics, motivations, barriers to participation, satisfaction, promotional effectiveness, and the relationship between social media engagement and in-person attendance.

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Capstone in Data Science
Fall 2025 · Fall 2026

V. Future: Predictive Modeling and Strategic Recommendations

This section explores opportunities to use historical and contemporary data to anticipate attendance patterns and develop actionable recommendations. It considers predictive modeling, improvements to longitudinal data collection, promotional strategies, and mechanisms for attracting and retaining student audiences.

VI. Conclusion

This section synthesizes the findings across the Past, Present, and Future phases, evaluates the project's contributions and limitations, and considers the implications of data-driven audience development for the Pomona College Music Department.